GEK-96873

DRIVE SYSTEM INSTRUCTIONS

RIG OPERATION MANUAL

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GE POWER SYSTEM
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RIG OPERATION MANUAL
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The information contained herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise that are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, Drive Systems Operations, Salem, Virginia, U.S.A.

Safety Symbol Legend

WARNING	Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in personal injury or death.
CAUTION	Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in damage to or destruction of equipment.
NOTE	Commands attention to an essential operating or maintenance procedure, condition, or statement that must be highlighted.

WARNING

THIS EQUIPMENT CONTAINS A POTENTIAL HAZARD OF ELECTRICAL SHOCK OR BURN. ONLY PERSONNEL WHO ARE ADEQUATELY TRAINED AND THOROUGHLY FAMILIAR WITH THE EQUIPMENT AND THESE INSTRUCTIONS SHOULD INSTALL, OPERATE OR MAINTAIN THIS EQUIPMENT.

ISOLATION OF TEST EQUIPMENT FROM THE EQUIPMENT UNDER TEST PRESENTS POTENTIAL ELECTRICAL HAZARDS. IF THE TEST EQUIPMENT CANNOT BE GROUNDED TO THE EQUIPMENT UNDER TEST, THE TEST EQUIPMENT'S CASE MUST BE SHIELDED TO PREVENT CONTACT BY PERSONNEL.

TO MINIMIZE HAZARD OF ELECTRICAL SHOCK OR BURN, APPROVED GROUNDING PRACTICES AND PROCEDURES MUST BE FOLLOWED.

CAUTION

THIS EQUIPMENT IS DESIGNED TO WORK IN A NORMAL INDUSTRIAL ATMOSPHERE. OPERATING RADIO TRANSMITTERS NEARBY MAY CAUSE THE CONTROL EQUIPMENT TO MALFUNCTION. IT IS RECOMMENDED THAT RADIO TRANSMITTERS NOT BE OPERATED WITHIN THE VICINITY OF THE CONTROL EQUIPMENT WHEN THE PANEL DOORS ARE OPEN, NOR AT ANY TIME WITHIN THE CONFINES OF A POWER CONTROL ROOM.

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CHAPTER 1
MAIN SWITCHBOARD

1-1. INTRODUCTION

The purpose of this instruction is to guide operating personnel on procedures to accomplish the following:

1. Operate the "bus tie" circuit breaker.
2. Start an engine.
3. Close a generator breaker on a dead bus.
4. Synchronize a generator to the bus.
   a. Manually
   b. Automatically
5. Set up a generator for "Blackout Start".
6. Removing a generator from service.
7. Check synchronizing frequency and voltmeters.

The operator should be familiar with operation of electrical power systems and with the switchboard one-line and front mounted devices as described in Figures 1-1, 1-1A, 1-1B, and Table 1-1. Normal system conditions are assumed. Procedures following a fault should be referred to maintenance personnel.

Instruction nameplates for key interlocking and the High Resistance Grounding (HRG) equipment are mounted on the switchboard and will not be repeated here.

1-2. BUS TIE CIRCUIT BREAKER

This circuit breaker can only be closed if one of the two busses (Bus A or Bus B) is dead.

Normal operation is with the "Bus Tie" circuit breaker closed. It is strictly manual and has no automatic trip devices.

Open or Close the breaker using the circuit breaker control switch (CB-CS) at the "Bus Tie" breaker compartment.

1-3. STARTING AN ENGINE

At the desired generator control panel:

1. Move the engine control switch (ENG-CS) to "idle" if engine is to start at idle speed. If permissible by the engine manufacturer, the engine may be started in the "Run" position. In "Run" the engine will accelerate to rated RPM.
2. Start the engine in accordance with the engine manufacturers instructions.
3. After a suitable time move the engine control switch to "Run".

1-4. CLOSING A GENERATOR BREAKER ON A DEAD BUS

At the synchronizing panel:

1. Set the synchronizing mode selector (MODE-CS) or "Off" or "Man Local".
2. Select the desired generator using the generator selector switch (GEN-CS).
3. Check incoming frequency (IN-Hz) for 60 Hz and incoming voltmeter (IN-VM) for 4160 V. Engine must be started and running at rated RPM. No adjustments are available.

At the generator control panel:

1. Move engine control switch (ENG-CS) to "Load" if not already in this position.
2. Close the circuit breaker with the breaker control switch(CB-CS).

At the engine control room panel:

1. Push the pushbutton to close the desired circuit breaker. (These pushbuttons are effective only on dead bus conditions.)

1-5. SYNCHRONIZING A GENERATOR TO THE BUS

1-5.1. Manual Synchronizing

1. Place synchronizing mode selector switch (MODE-CS) in "Man Local".
2. Place generator synchronizing selector switch (GEN-CS) in G1, G2 or G3 as appropriate.
3. Start engine.
4. Place engine control switch in "Load" position.
5. Check incoming, bus frequency, and voltmeters. No adjustments should be required.
6. Move the synchronizing lights (SL) are visible from the selected generator panel.
7. The synchroscope should be rotating slowly in either direction. As the pointer approaches "0", close the circuit breaker using the breaker control switch at the generator panel.
   engine speed adjustment is required unless the synchroscope is stationary and more than 10-15 degrees from "0". If necessary, a slight adjustment in engine speed can be made using the speed trim adjustment rheostat (SPD).
   A synchronizing check device prevents closing if

voltage and phase relationships are not within acceptable limits.

Once the breaker closes, load sharing becomes automatic and the speed adjustment rheostat has little or no effect.

Turn the synchronizing selector switch "off".

1-5.2. Automatic

1. Place mode selector switch in "Auto-Local".
2. Place generator selector switch in the desired position (G1, G2, or G3).
3. Start engine and place engine control switch in "Load". When all conditions are satisfied, the breaker closes automatically.

1-6. BLACKOUT START

1. Place synchronizing mode selector switch (MODE-CS) in "Standby".
2. Set the engine control switch on the generator to be brought on line automatically after a blackout in the "Load" position.
3. Place the "Standby Selector Switch" located in the engine control room in "G1", "G2, or "G3" as desired. The corresponding "standby" light will glow to confirm that the generator selected is in the automatic start mode.

   The generator selected by the remote "Standby" selector switch will start and automatically connect to a dead bus after a blackout when voltage and frequency are within limits.

1-7. REMOVING A GENERATOR FROM SERVICE

1. Insure that the load shed by removing a generator from service can be accommodated by the generators left in service. One way to determine this is by comparing the two KW meters in the wheelhouse.
2. Two procedures are available to open a circuit breaker:

   a. Unloading engine before opening CB.

      - At the generator panel move the engine control switch from "Load" to "Run". This transfers load from the generator to be removed to the remaining generators while the generator CB is still closed. If the remaining generators cannot accept the added loading, the engine control switch can be quickly moved back to "Load" without the necessity of re-synchronizing.

      - Open the circuit breaker using the breaker control switch or the "Open" pushbutton at the engine control room station.

   b. Opening CB immediately

      - Open the circuit breaker using the breaker control switch or the "Open" pushbutton at the engine control room station.

1-8. CHECKING SYNCHRONIZING FREQUENCY AND VOLT METERS

1. Move the synchronizing calibrate switch (STST) to the "Sync Calibrate" position and observe that both frequency meters read the same and that both voltmeters read the same.

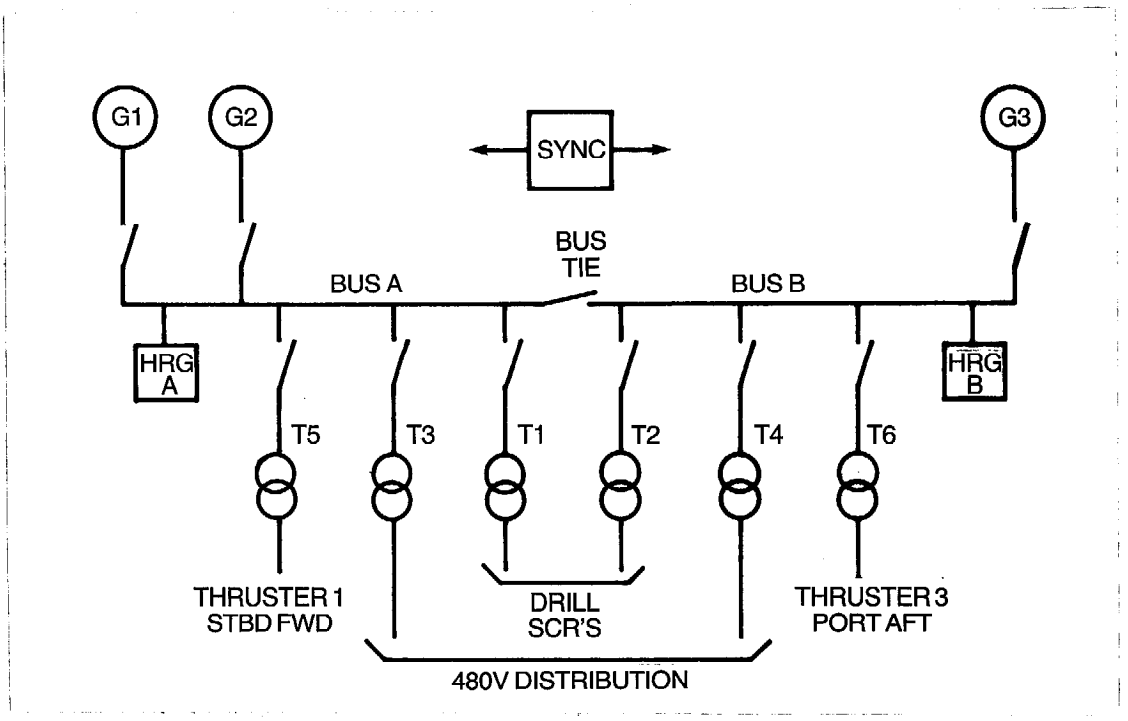


Figure 1-1.
Main Switchboard One-Line

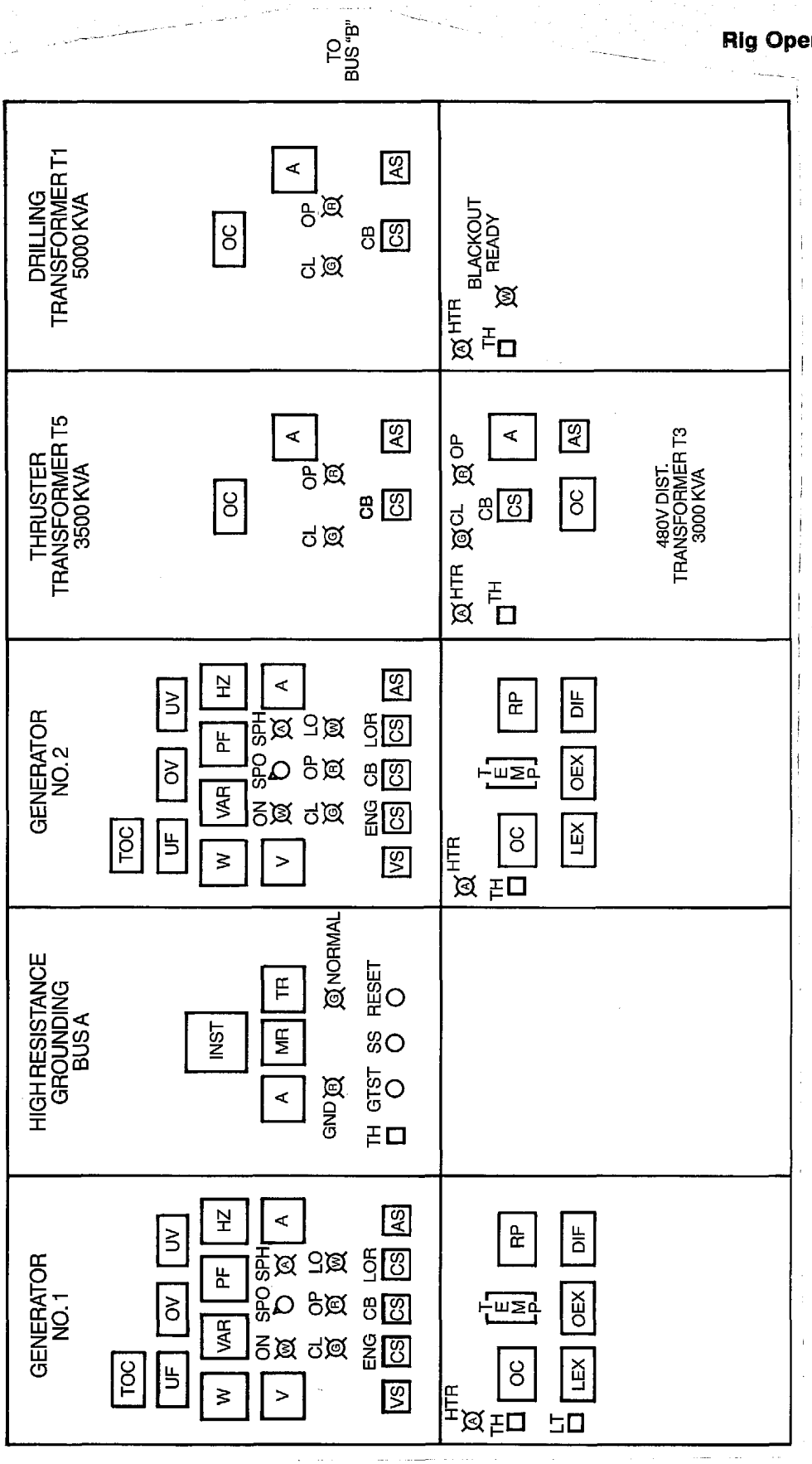


Figure 1-1A.
Main Switchboard Front View
Bus "A"

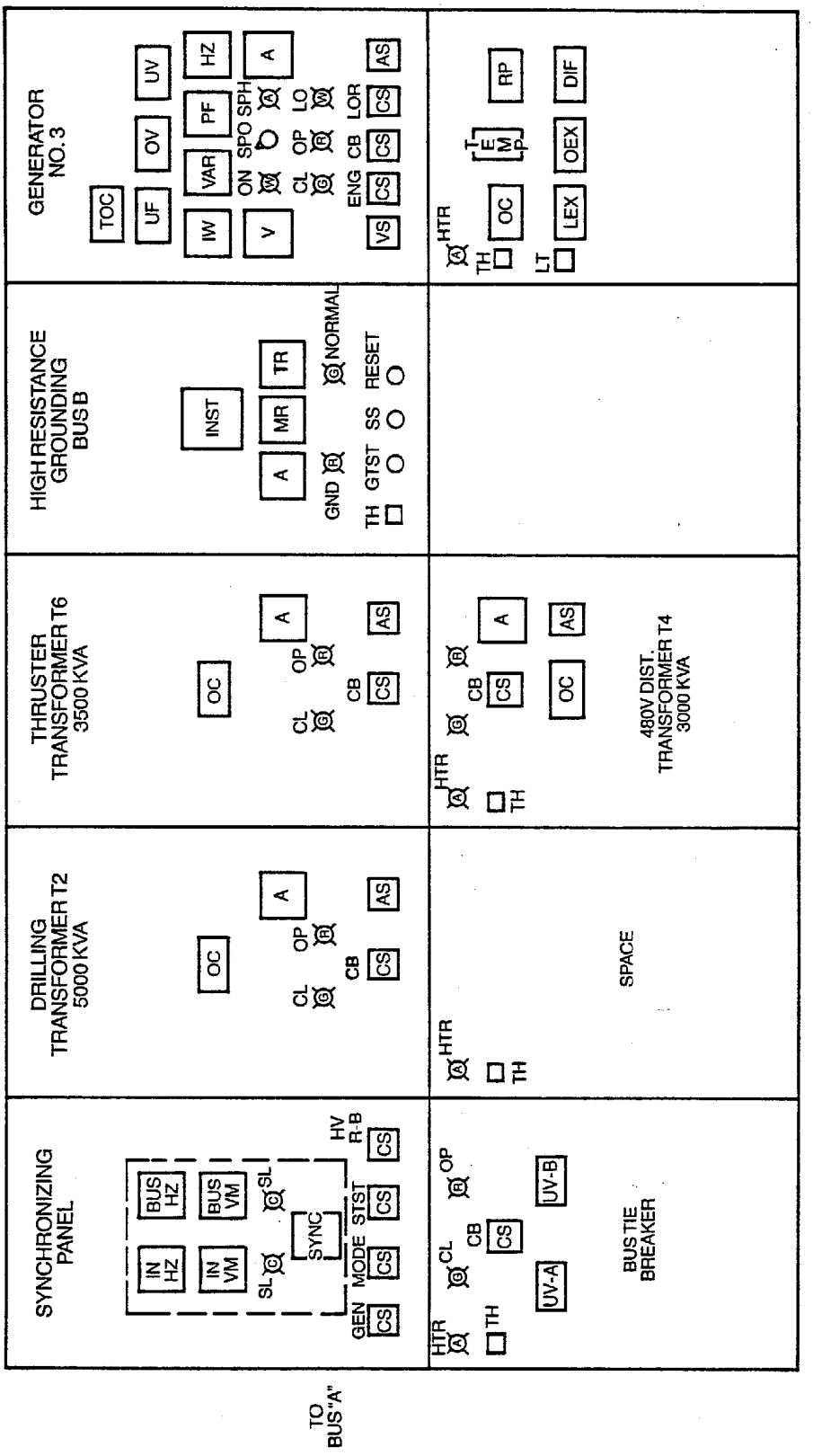


Figure 1-1B.
Main Switchboard Front View
Bus "B"

Table 1-1.
Main Switchboard Front View Mnemonics

MNEMONIC	DESCRIPTION
A	AMMETER
AS	AMMETER PHASED SWITCH
BUS HZ	BUS FREQUENCY METER
BUS VM	BUS VOLTMETER
CB	CIRCUIT BREAKER
CL	CIRCUIT BREAKER CLOSED LIGHT
CS	CONTROL SWITCH
DIF	DIFFERENTIAL RELAY
ENG	ENGINE
GEN	GENERATOR SELECTOR SWITCH (SYNCHRONIZING)
GND	GROUND FAULT LIGHT
GTST	GROUND FAULT TEST SWITCH
HTR	PANEL SPACE HEATER ON LIGHT
HV A-B	SYNCHRONIZING BUS SELECTOR SWITCH
HZ	FREQUENCY METER
IN HZ	INCOMING GENERATOR FREQUENCY METER
IN VM	INCOMING GENERATOR VOLTMETER
INST	INSTRUCTIONS FOR GROUND DETECTOR
LEX	LOSS OF EXCITATION RELAY
LO	LOCKOUT RELAY OK LIGHT
LOR	LOCKOUT RELAY (INHIBITS VOLTAGE REGULATOR)
LT	LIGHT SWITCH (FLUORESCENT LIGHTS UNDER DRIPSHIELD)
MODE	SYNCHRONIZING MODE SELECTOR SWITCH
MR	METER RELAY
NORMAL	NO GROUND FAULT LIGHT
OC	OVERCURRENT RELAY
OEX	OVEREXCITATION RELAY
ON	GENERATOR ON INDICATING LIGHT
OP	CIRCUIT BREAKER OPEN LIGHT
OV	OVERVOLTAGE RELAY
PF	POWER FACTOR METER
RESSET	GROUND FAULT RESET PUSHBUTTON
RP	REVERSE POWER RELAY
SL	SYNCHRONIZING LIGHTS
SPD	ENGINE SPEED ADJUSTING RHEOSTAT FOR SYNCHRONIZING
SPH	GENERATOR SPACE HEATER ON LIGHT
SS	GROUND DETECTOR MODE SELECTOR SWITCH
STST	SYNCHRONIZING METER TEST SWITCH
SYNC	SYNCHROSCOPE
TEMP	RTD TEMPERATURE TRANSDUCER
TH	PANEL SPACE HEATER THERMOSTAT
TOC	OVERCURRENT RELAY WITH TIME DELAY
TR	TIMING RELAY
UF	UNDERFREQUENCY RELAY
UV	UNDERVOLTAGE RELAY
UV-A	BUS A UNDERVOLTAGE RELAY
UV-B	BUS B UNDERVOLTAGE RELAY
V	VOLTMETER
VAR	VAR METER
VS	VOLTMETER PHASE SWITCH
W	WATTMETER

CHAPTER 2
EMERGENCY GENERATOR CONTROL

2-1. EMERGENCY GENERATOR CONTROL PANEL

The emergency generator control panel is a combination of engine/generator control and power distribution to those electrical loads classed by regulatory bodies as critical for vessel operation. The ship service power distribution panel (provided by others) supplies the normal power source for these electrical loads.

2-1.1. Emergency Power Distribution

The emergency power distribution panel consists of standard motor control center units, including both motor starters and feeder circuit breakers. During normal operation, it receives its power from the ship service power distribution panel through the tie circuit breaker. This power feeds the emergency panel supply bus. When the normal power supply fails, an engine/generator set supplies the emergency electric power source.

When the emergency panel's controls are set up for manual/automatic operation (switches in MANUAL and AUTOMATIC), the tie circuit breaker can be closed and the generator circuit breaker is open. If the normal electric power source fails, the tie circuit breaker's undervoltage trip unit opens the tie circuit breaker. This sends a signal to the engine control panel to start the engine. As soon as the engine reaches operating speed, the emergency panel's voltage regulator starts controlling the generator output. When the generator voltage approaches the normal 480 V, the generator circuit breaker automatically closes, supplying an emergency source of electric power to the emergency distribution panel.

The emergency generator control panel contains volt, ampere, kilowatt, and frequency meters to indicate generator output.

When in the "AUTOMATIC" mode normal source of vessel power (from the ship service distribution panel) is re-established to the emergency panel's tie circuit breaker, voltage sensing circuits trip the emergency generator circuit breaker and immediately close the tie circuit breaker. Then vessel electric power distribution is back to normal.

The emergency engine continues to run for approximately five minutes at the "no load" condition to permit a normal cool down cycle. After five minutes, the engine is automatically shut off and all circuits return to normal, until the next power failure from the ship service distribution panel.

2-1.2. Manual Mode of Operation (Testing)

For testing the emergency generator function, the emergency panels include manually controlled switches and buttons. The operator can put the engine/generator into a manual mode with the Manual-Automatic selector switch, then can start the engine. In this mode, the operator can trip the tie circuit breaker and manually initiate closing of the generator breaker.

CHAPTER 3
DRILLING/AW SYSTEM OPERATION

3-1. INTRODUCTION

This chapter describes the location, purpose, and operation of the µ drill™ 3000 drive system controls and indicators. The µdrill™ 3000 is a modular rig drive system with individual SCR bridges for each motor.A typical rig drive system includes a driller's console, power control equipment (often mounted in a power control room), a mud pump console (optional), and the necessary motors and generators. Figure 3-1 is a simplified diagram of a typical µdrill™ 3000 rig drive system.

The **driller's console** contains controls such as selection switches (used to enable motor drives) and motor throttles. Indicators show equipment status such as motor load, SCR status, generator status, plus the status of oiler and blower motors.

The **mud pump console**, when used, contains a throttle and status lights for each mud pump. A switch on the driller's console determines whether the driller's console or mud pump console has control of the mud pump motors.

The **µdrill™ 3000 control** equipment contains a variety of controls and indicators, and is available in a variety of configurations. There will be differences between rigs in such things as the number of motors, generators, and SCRs, the control functions available, whether the power control equipment is mounted in a power control room, and the types of options installed. Because of this, your rig drive system may not look exactly like some of the photographs or diagrams in this book. In addition, there may be descriptions of equipment or controls that your system does not have. Keep this in mind while using this manual, since all of the information may not apply to your particular rig.

3-2. RIG CONTROLS AND INDICATORS

On the next several pages are photographs showing controls and indicators for the driller's console, mud pump console, and µdrill™ 3000 control equipment. Controls pointed out in the photograph are described in nearby text. See Chapter 3 for rig operating procedures.

3-2.1. DRILLER'S CONSOLE

The indicators, meters, throttles and switches on the driller's console give the driller control over the drilling

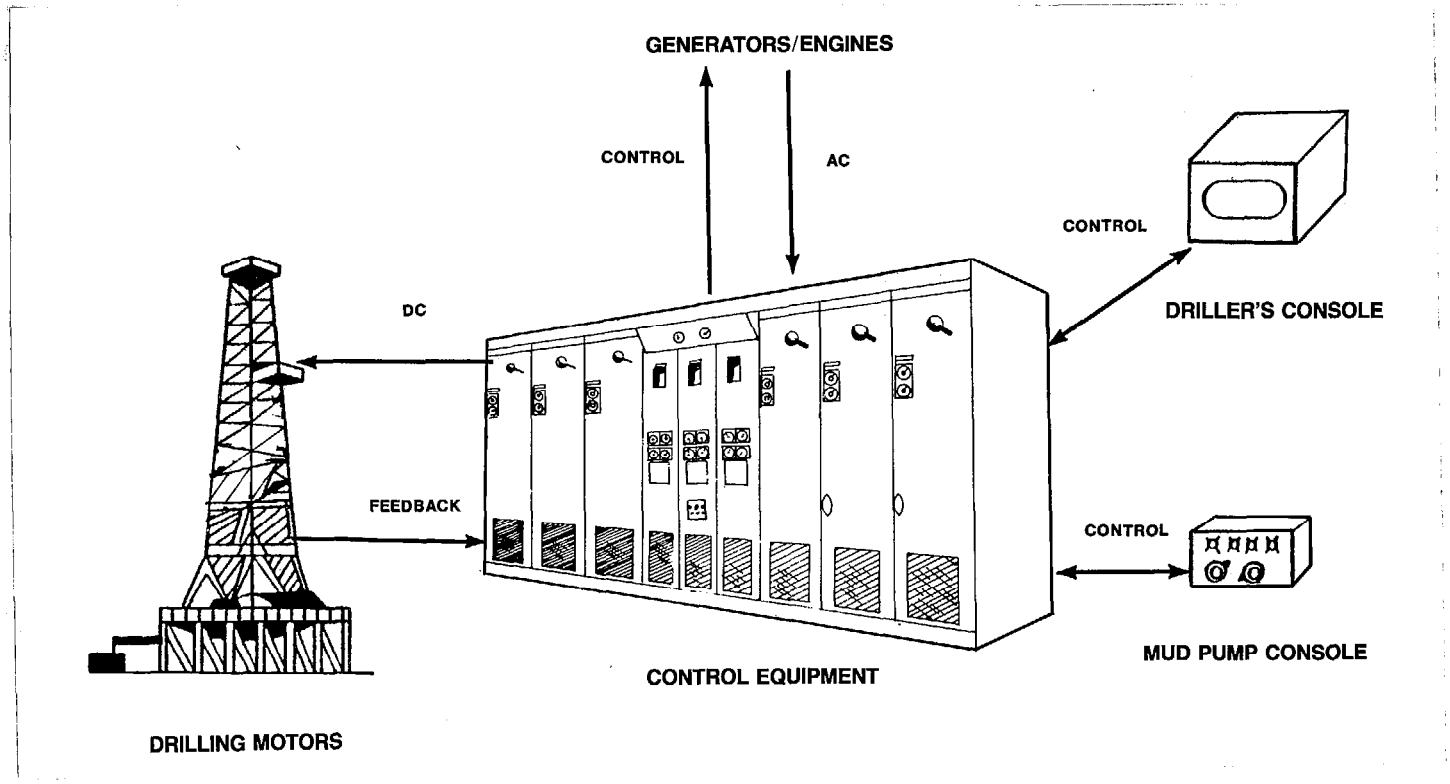


Figure 3-1.
Typical Rig Drive System

motors. A description of each item on the driller's console is given below.

3-2.1.1. STATUS LIGHTS. These lights show the operating condition of rig equipment such as generators, SCRs, and motors. Each light is labeled with the name of the equipment or function monitored. When the light is on, the equipment or function is active. Some typical items that may be monitored by the status lights are listed below.

Gen On -The indicated generator is supplying power to the drive system when this light is on.

Drive Ready -The drive is selected and ready to run when this light is on. When the light flashes, it indicates that the drive is selected and awaiting permissives or indicates a warning if the drive is running.

Supercharger On -The supercharger is running when this light is on. When the light flashes, it indicates the supercharger failed to start.

Auxiliaries -The mud pump auxiliary motors are running when this light is on. When the light flashes, it indicates the auxiliary motors failed to start.

High Load -The engine/generator sets supplying power to the rig are approaching an overload condition when this light comes on.

Alarm Silenced -The alarm is silenced when this light is flashing. In this condition, the audible alarm will not sound for abnormal conditions.

3-2.1.2. SELECTION SWITCHES. These switches allow the driller to select specific drilling motor functions and directions. The number of selection switches depends on the number of drilling motor functions in the rig drive system.

Mud Pump Switches -A mud pump is selected by turning the desired switch (MP1 or MP2) to the ON position. Control of the mud pump is transferred to the mud pump console by turning the switch to the REM position.

Drawworks Switch -The drawworks are enabled when the DW switch is in either the FWD or REV position.

Rotary Table Switch -The rotary table is enabled when the RT switch is in either the FWD or REV position.

3-2.1.3. THROTTLES. The throttles control the speed of the motors indicated on the respective console nameplates. The number of throttles depends on the number of drilling functions in your particular configuration. A typical configuration would include four throttles: one for mud pump 1, one for mud pump 2, one for the drawworks, and one for the rotary table.

3-2.1.4. ROTARY TABLE TORQUE LIMIT CONTROL. This control limits the torque provided to the rotary table. The rotary table may be chain-driven by the drawworks motor or it may be an independent rotary with its own motor.

3-2.1.5. EMERGENCY SHUTDOWN. Pressing this button trips the SCRS disconnect, thus removing motor field and armature power. ***Other power may still be connected to the motor at his time!***

3-2.1.6. POWER PLANT SHUTDOWN. Pressing this button shuts off the field supply to the engine/generator sets, thus tripping the generator breakers. The button is mounted under a spring-loaded plastic cover to prevent accidental activation (optional). ***Complete shutdown of the engine/generator sets may require an air supply cut-off mechanism.***

3-2.1.7. %KW METER. This meter indicates how much of the total power available from the engines is being used by the rig. Using this meter, an operator can determine the correct number of engines to run to supply enough power for rig operation. The position of the needle indicates the number of engines required.

3-2.1.8. Rotary Table Torque Meter. This meter indicates the amount of torque applied to the drill pipe by the rotary table. The meter may be calibrated both in foot-pounds and amperes.

3-2.1.9. ALARM SILENCE SWITCH. When activated, this switch prevents the audible alarm from sounding at the driller's console. The alarm silenced light will flash as long as this switch in ON.

3-2.1.10. AUDIBLE ALARM. Unless disabled by the alarm silence switch, the audible alarm sounds in case of:

1. Loss of a motor blower while the motor is running.
2. Loss of the supercharger while a mud pump motor is running.
3. Loss of a mud pump auxiliary motor (i.e., a lube motor) while the mud pump motor is running.

3-2.2. Mud Pump Console

The mud pump console is simply an extension of the driller's console and is used to control the mud pumps from a remote location.

3-2.2.1. THROTTLES. The mud pump throttles serve the same purpose on the mud pump console as on the driller's console: Controlling the speed of the mud pump motors.

3-2.2.2. INDICATORS . There are normally four indicators on the mud pump console, one for each mud

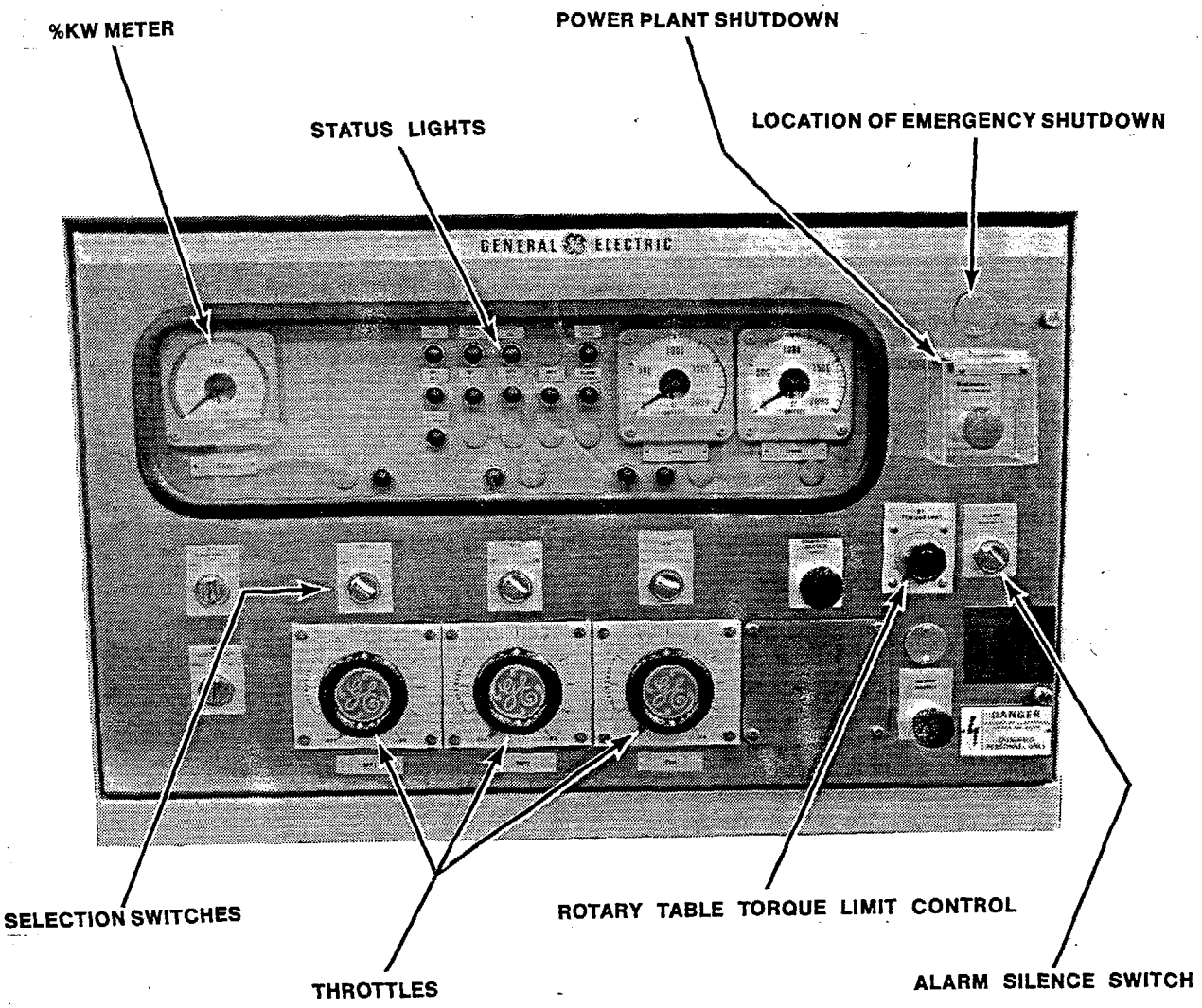


Figure 3-2.
Typical Driller's Console

pump motor. These indicators are labeled DRIVE READY HERE and will light only if the following conditions are met:

1. The remote throttle must be selected by turning the mud pump selection switch on the driller's console to the REM position.l
2. The auxiliary and supercharger motors must be operating.
3. The power converter must be operating.

3-2.2.3. POWER CONVERSION PANEL. The power conversion panel uses SCRs (Silicon Controlled Rectifiers) to convert AC voltage supplied by the generators to a variable DC voltage for the drilling motors. As shown in Figure 3-4, the front of this panel contains two indicators, two meters, and a reset button that monitor the operation of the SCRs:

Voltmeter -This meter indicates the DC voltage supplied to the drilling motor by the SCR.

Ammeter -This meter indicates the DDC current supplied to the drilling motor by the SCR.

SCR OK Light -When ON, indicates that the SCR is available for drive selection.

Drive Ready Light -When on, indicates that the drive is ready to run. When off, indicates that the drive is not selected. When flashing, indicates that the selected drive is awaiting permissives or indicates a warning if the drive is running.

Reset Pushbutton -Pressing this button causes the power conversion panel to check itself for shorted cells or failure of a cell to fire. This button is active only when the drive is not selected.

3-3. Operating Procedures

The procedures outlined below are for normal, day-to-day operation of the μdrill™ 3000 rig drive system, including rig start up, operation, and shut down.

These procedures assume a rig drive system with typical options installed. Your system may not have some of the controls or indicators mentioned. In addition, rig drive systems vary in the number of generators, generator controls, power control panels, and motors. Disregard any steps that do not apply to your rig drive system.

In each of the steps below, the names of modules, panels, indicators, and controls which are described in this manual are printed in bold type. For descriptions of other devices, refer to the appropriate vendor publications.

3-3.1. Drilling System Operation

Once the rig drive system has been started, the drilling motors are controlled from the driller's console and

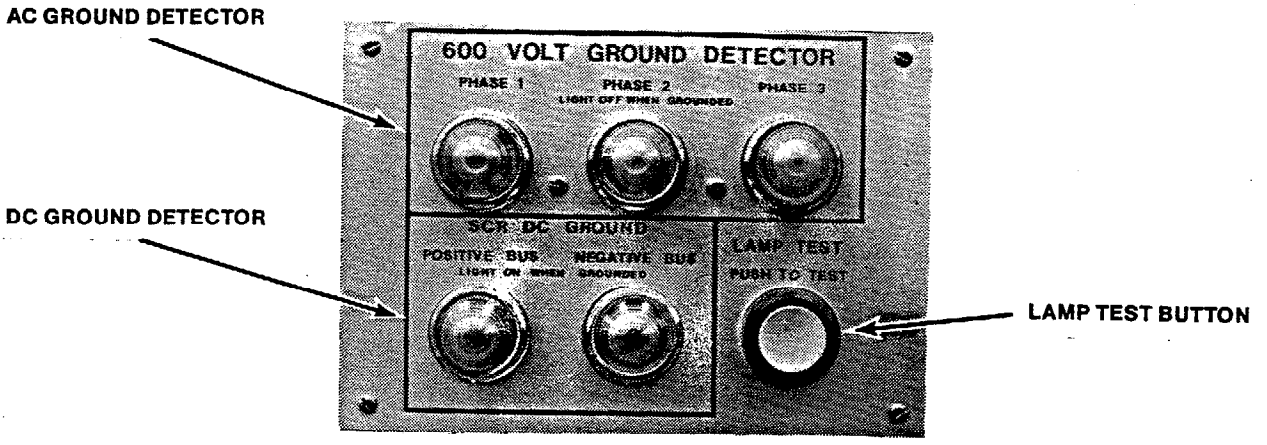


Figure 3-3.
Ground Detector Module

mud pump console. The operator will seldom need to concern himself with the control equipment.

Most operations can be broken down into four types of activities:

- Making motor selections (including alternate assignments)
- Controlling motor speed and direction
- Monitoring rig operation
- Correcting operating problems

These activities are described below. As we describe each activity, refer back to the photograph of the driller's console (Figure 3-2).

3-3.1.1. MAKING MOTOR SELECTIONS . The selection switches on the driller's console are used to enable SCR power converters for various drilling functions and directions. Each selection switch represents a specific drilling function; for example, mud pump #1, the drawworks, or the rotary table.

Once the motor has been selected, the DRIVE READY light for that motor will come on and the motor can be controlled with the throttle. If the rig has a purge option, the DRIVE READY light will flash for six seconds before coming on. If the light flashes for more than six seconds, one or more of the following problems exist:

- The motor blower may have failed to start or the signal from it may have failed to reach the power conversion panel.
- The throttle at the driller's console may not be in the OFF position.
- The field or reversing contactor may have failed.
- An auxiliary drive motor (optional) may have failed to start.

CAUTION

If the DRIVE READY light begins to flash during drilling, a problem has arisen. Continued operation under these conditions may cause damage to equipment.

Troubleshooting of this equipment is covered in the μdrill™ 3000 Troubleshooting Guide.

3-3.1.2. ALTERNATE ASSIGNMENT (OPTIONAL) . If your rig contains the alternate assignment feature, there will be an alternate assignment selection switch on the driller's console. The alternate assignment option allows the driller to use a single power conversion panel for more than one drilling function. The most common alternate assignment is to use the drawworks "B" power conversion panel to power the drawworks "B" motor

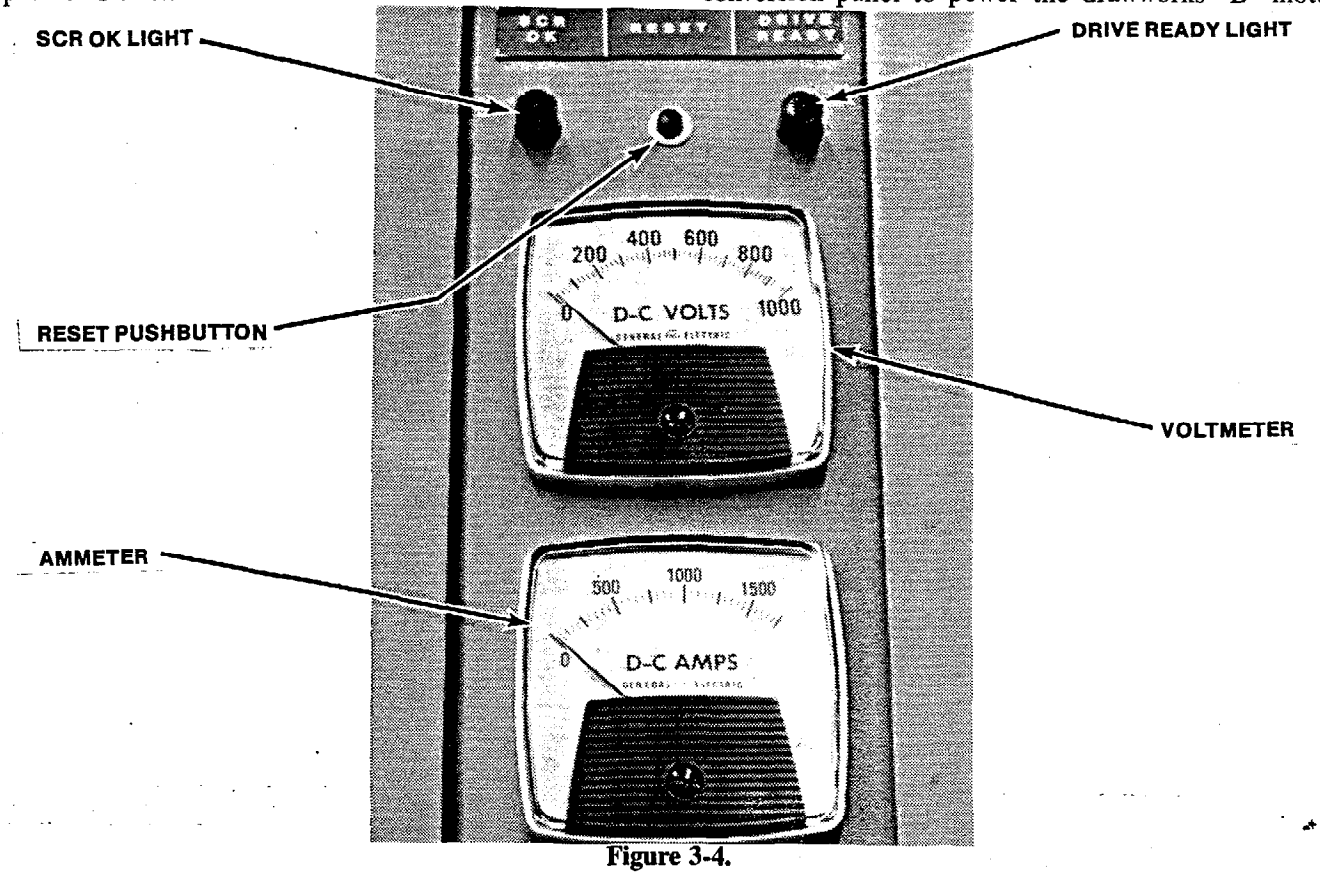


Figure 3-4.
Power Conversion Panel Indicators

(normal) or the rotary table motor (alternate assignment). Other alternate assignments are available but all typically operate the same.

Using the drawworks "B" alternate assignment option described above as an example, a typical alternate assignment operation is as follows. In the normal mode, the drawworks "B" power conversion panel will supply power to the drawworks "B" motor. In the alternate assignment mode, the drawworks "B" power conversion panel will supply power to the rotary table and the drawworks "B" motor will be disabled. The mode of operation is selected using the alternate assignment selection switch on the driller's console. After selection of alternate assignment all controls, indicators, and auxiliaries will operate normally for the alternate motor.

3-3.2. Controlling Motor Speed and Direction

The driller's console is the main control center for all rig motors. The driller can control both motor speed and direction with the motor throttles and selection switches.

3-3.2.1. MUD PUMP. The driller selects a mud pump by turning the proper selection switch to the ON position. The pump is then controlled using the corresponding throttle. If the rig has an optional mud pump console, the driller can transfer control of the mud pump to that console by turning the selection switch to the REM position. With the selection switch in the REM position, the DRIVE READY lights on the mud pump console will light. If these lights begin flashing, there is a problem (as described above under "Making Motor Selections") that should be corrected. If the lights illuminate continuously, the mud pumps are ready to run.

Under normal operating conditions, two motors on the mud pump share the pumping load. If one-motor operation is required, the motor not in use should be de-assigned at the power control room. When this is done, the µdrill™3000 drive system will automatically recalibrate the current limit on the remaining motor, allowing that motor to supply more power than it would under two-motor operation.

3-3.2.2. DRAWWORKS. The drawworks are controlled from the driller's console only. The driller controls the drawworks using a single selection switch and single throttle. The selection switch enables operation of the motor in either the forward (FWD) or reverse (REV) direction. The FWD position of the switch enables regenerative braking of the drawworks motors. Regenerative braking is a method of slowing down a motor by pumping its energy back into the AC power system, thus allowing the motor to decelerate more rapidly.

Drawworks motor speed is controlled by either a hand or foot throttle at the driller's console. Usually, cathead speed is set with the hand throttle and hoist speed is controlled with the foot throttle. Deselection of the drawworks motor must be done in the power control room.

3-3.2.3. ROTARY TABLE. The rotary table is driven by its own motor, which is controlled by a selection switch and throttle on the driller's console. The rotary table torque limit control is used to limit the torque applied to the drill string. This torque limit also limits cathead torque. The µdrill ™ 3000 drive system is the first rig system to allow the driller to control cathead torque.

The rotary table may be chain-driven and clutched off the drawworks motors. In this case, the driller uses the drawwork selection switch and throttle to control the rotary table. Clutching determines the correct current limit value for drawworks or rotary table operation. Whenever the drawworks is unclutched, the rotary table torque limit controls cathead and rotary torque.

3-3.3. Monitoring Rig Operation

There are several indicators and meters on the consoles that should be monitored during rig operation. Make a habit of occasionally scanning the console indicators and meters to ensure proper rig operation and early detection of problems.

Should any of the status lights show an abnormal condition the audible alarm will sound, unless the alarm silence switch is ON. Leave the silence switch OFF unless you need to silence an active alarm while you fix a problem.

Before starting a drilling operation or making a function change, the driller should check the following things:

1. Are the correct number of GEN ON lights ON?
2. Are all selections switches set properly and are the correct DRIVE READY lights on?
3. Are all throttles in the OFF position?
4. Is the rotary table torque limit control set according to the drill pipe grade and diameter?
5. Is the drawworks selection switch set in the correct direction (FOR or REV)?

3-3.4. Correcting Operating Problems

Two common operating problems are described below along with suggestions for correcting them:

Insufficient Rig Power - If the speed of all DC motors decreases and the KVA/KW limit light comes on, there are not enough engines/ generators on line to support the heavy drilling load. Should this occur, the load must be reduced or an additional generator brought on line.

Insufficient RT Torque - If only the rotary table's speed decreases, the torque limit set by the rotary table torque limit control has been reached. Ensure that the limit

control is set properly for the type of pipe being used. If the setting is correct, you must reduce the torque requirements. There may be too much weight on the bit or the wrong gear may have been selected.

CHAPTER 4
THRUSTER POWER CONVERTERS

4-1. Thruster Control

The thrusters can be controlled from four different locations; the wheel house, the thyristor converter panel, the thruster motor compartment, or the engine control room.

1. The engine control room station is the principal control location. Each thruster will have a control location switch located at this station and will assign control to this station, the wheel house, the SCR panel, or the motor compartment. When control is transferred to SCR panel, wheel house, or motor compartment, the receiving station will have to acknowledge before control is actually transferred. The engine room control station can take control away from any of the other stations without warning. Pushing the "take control acknowledge" pushbutton will acknowledge and transfer control.
2. The engine room thruster control station selector switch will have positions to choose either "Off", "Local", "Wheel House", SCR panel or motor compartment. In "Local", the thruster motor speed will be determined by the local speed throttle. In "Wheel House" position, the Wheel House throttle will control all thrusters that are so assigned and on line to operate at the same motor horsepower. Azimuth control in all cases is by the thruster vendor.
3. The wheel house control station will also include indication of how many generators are on line and a percent KW meter that will indicate total system load. The system high load light will indicate that the system is loaded to about 95% of on line capacity.
4. The SCR panel and motor compartment thruster control stations will have the same devices.

Thruster controls will be arranged in four groups, one for each control location.

The control location switch on the engine room control station (one per thruster) assigns thruster control to the wheel house, the SCR panel controls, or the thruster motor compartment controls.

Each thruster control switch includes the following four positions:

- Off - That specific thruster is not assigned to operate. All auxiliaries controlled from the thruster SCR panels will be off.
- Engine Control Room - The thruster auxiliaries will be turned on and the SCR converter will be ready to operate if all permissives are OK. Thruster HP reference will come from devices supplied by the thruster vendor. A separate reference is provided for each thruster.
- Wheel House - The thruster is ready to go if all permissives are OK. Thruster HP reference will come from devices supplied by the thruster vendor. A separate reference is provided for each thruster.
- Thruster SCR Panel
- Thruster Motor Room

Operation of the thruster SCR is similar to the drilling SCRs. Refer to KaMeWa instruction book for further information.

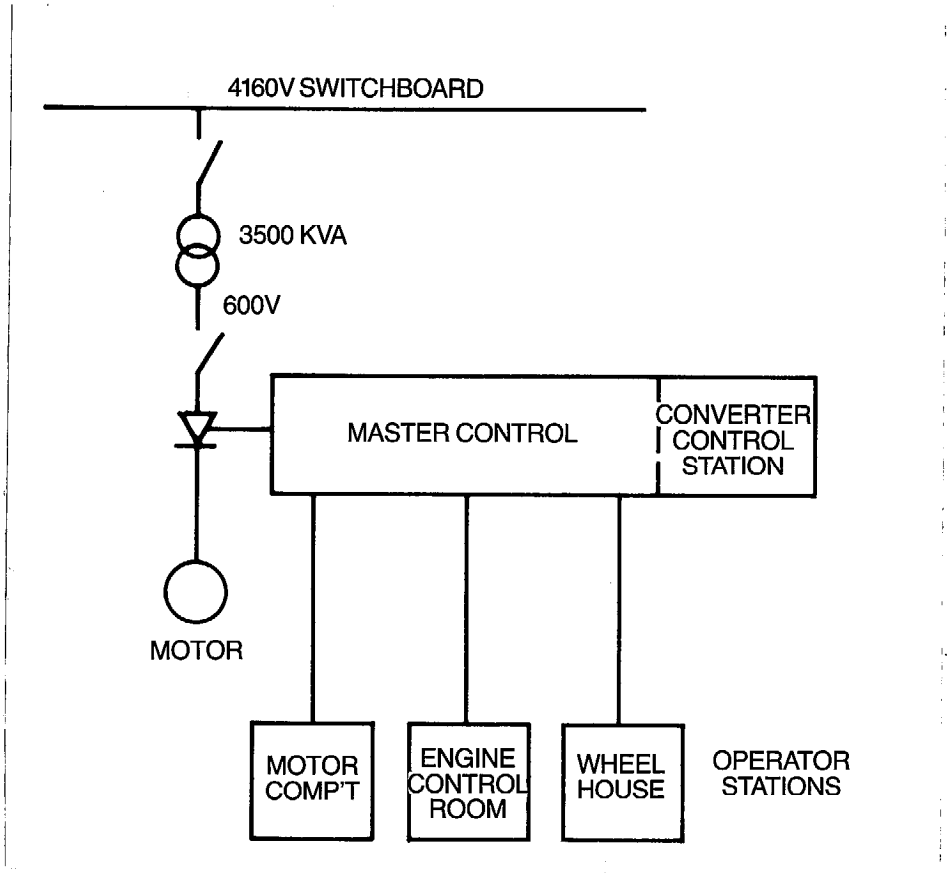


Figure 4-1.
Thruster Control Block Diagram

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